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High Level Programming Extensions – Function

Problem 1:

Create a function that returns the level of a customer based on credit limit.

(Use the IF statement to determine the credit limit).

If credit limit > 50000 then customer_level = PLATINUM

If credit limit >= 10000 AND credit limit <= 50000 then customer_level = GOLD

If credit limit credit limit < 10000 then customer_level = SILVER

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>  create database collegedb;
Query OK, 1 row affected (0.06 sec)

mysql> use collegedb;
Database changed
mysql> create table customers( customerNumber int primary key, customerName varchar(30), creditLimit decimal(10,2) );
Query OK, 0 rows affected (0.24 sec)

mysql> insert into customers values (101,'Ramesh',75000), (102,'Pranathi',45000), (103,'Sonu',8000);
Query OK, 3 rows affected (0.14 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from customers;
+----------------+--------------+-------------+
| customerNumber | customerName | creditLimit |
+----------------+--------------+-------------+
|            101 | Ramesh       |    75000.00 |
|            102 | Pranathi     |    45000.00 |
|            103 | Sonu         |     8000.00 |
+----------------+--------------+-------------+
3 rows in set (0.05 sec)

mysql> delimiter //
mysql> create function customer_level(climit decimal(10,2)) returns varchar(20) deterministic begin declare level_name varchar(20); if climit > 50000 then s
et level_name = 'PLATINUM'; elseif climit >= 10000 and climit <= 50000 then set level_name = 'GOLD'; else set level_name = 'SILVER'; end if; return level_na
me; end //
Query OK, 0 rows affected (0.06 sec)

mysql>  delimiter ;
```

```
mysql> select customer_level(75000) as Customer_Level;
+----------------+
| Customer_Level |
+----------------+
| PLATINUM       |
+----------------+
1 row in set (0.06 sec)

mysql>  select customer_level(45000) as Customer_Level;
+----------------+
| Customer_Level |
+----------------+
| GOLD           |
+----------------+
1 row in set (0.05 sec)

mysql>  select customer_level(8000) as Customer_Level;
+----------------+
| Customer_Level |
+----------------+
| SILVER         |
+----------------+
1 row in set (0.00 sec)
```

Problem 2

Write a recursive MySQL procedure compute the factorial of a number.

```
mysql> create database collegedb;
Query OK, 1 row affected (0.10 sec)

mysql>  use collegedb;
Database changed
mysql> set @@session.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.05 sec)

mysql> delimiter //
mysql> create procedure factorial( n int, out result bigint ) begin if n <= 1 then set result = 1; else call factorial(n - 1, result); set result = result *
 n; end if; end //
Query OK, 0 rows affected (0.07 sec)

mysql> delimiter ;
mysql> set @result = 0;
Query OK, 0 rows affected (0.05 sec)

mysql> call factorial(5,@result);
Query OK, 0 rows affected (0.05 sec)

mysql> select @result as Factorial;
+-----------+
| Factorial |
+-----------+
|       120 |
+-----------+
1 row in set (0.00 sec)
```